10-11-2026

Page
Date

CO4 - Practical Exercises

4. Threat Intelligence Application Area Identifier

Pseudocode:

1. Start.
2. Read the business scenario.
3. Convert the scenario to lowercase.
4. Define Keywords for banking, healthcare, e-commerce, cloud, and telecommunication security.
5. Match the scenario with the Keywords of each application area.
6. Calculate the matching score for each area.
7. Select the area with the highest score.
8. Display the most relevant application area.
9. Stop.

Sample Input:

A bank wants to detect suspicious credit card transactions and prevent account takeover attacks.

Sample Output:

Most relevant Threat Intelligence Application Area: Banking Fraud Prevention.

2. Vulnerability Assessment Using a Port Scanner

Pseudocode:

1. Start
2. Read the authorized target hostname or IP address.
3. Read the starting and ending port numbers.
4. Create a TCP socket for each port.
5. Attempt to connect to each port.
6. If the connection succeeds, display the port as OPEN.
7. Repeat until all specified ports are checked.
8. Display scanning completion.
9. Stop.

Sample Input:

Enter target host = 127.0.0.1
Enter starting port = 80
Enter ending port = 82

Sample Output:

Open Ports: Port 80 is OPEN
Port scanning completed.

3. Security Policy Governance Checker

Pseudocode:

1. Start
2. Read the security policy document.
3. Convert the document to lowercase.
4. Define the required elements: scope, roles and responsibilities, enforcement, review cycle and approval authority.
5. Check whether each required element is present.
6. Store the missing elements.
7. If any elements are missing, display them.
8. Otherwise, display that all required elements are present.
9. Stop.

Sample Input:

The scope covers all employees.
Roles and responsibilities are assigned.
END.

Sample Output:

Missing Governance Elements:
- Enforcement
- Review Cycle
- Approval Authority

4. VPN Confidentiality Demonstration using a Symmetric Stream Cipher

Pseudocode:

1. Start
2. Read the original message and shared key
3. Convert the message and key into bytes.
4. Encrypt the message using a simple XOR cipher.
5. Display the encrypt the data using a wrong key.
6. Decrypt the data using the correct shared key.
7. Display the decrypted message
8. Stop.

Sample Input:

Enter message: Hello
Enter shared key: Key

Sample Output:

Captured data (Hex): 230015070a
Decrypted with wrong key: unreadable or incorrect data
Decrypted with correct key: Hello

5. Threat Intelligence Report Classifier

Pseudocode:

1. Start
2. Read the threat-intelligence report description.
3. Read the intended audience.
4. Convert both inputs to lowercase
5. Define keywords for Strategic, Tactical and Operational intelligence.
6. Match the inputs with the defined keywords.
7. Calculate the score for each category.
8. Select the category with the highest score.
9. Display the classification.
10. Stop.

Sample Input:

Enter report description: Long-term cyber risk trends and business impact.
Enter intended audience: Executives.

Sample Output:

Threat Intelligence Classification: Strategic

6. Phishing URL Detection

Pseudocode

1. Staxt
2. Read the URL
3. Paxse the URL and extxact the hostname.
4. Check fox suspicious keywoxds.
5. Check whethex the hostname is an IP addxess
6. Check whethex the URI contains excessive subdomains.
7. Recoxd all detected suspicious indicatoxs.
8. If indicatoxs axe found, display a waxning.
9. Othexwise, display that no obvious phishing indicatoxs wexe found.
10. Stop.

Sample Input :

Entex URL : http://192.168.1.10/login

Sample Output:

Waxning: Suspicious URL indicatoxs found:
- Suspicious keywoxd detected
- IP address used as domain

1. ELK Stack Simulation Using Python

Pseudocode:

1. Start
2. Read the raw log entries
3. Parse each log into its level and message (Logstash-style processing)
4. Store the structured events in an index
5. Count the events by log level
6. Display the indexed events
7. Display the summary of log levels
8. Stop.

Sample Input:

INFO User login successful
ERROR failed login attempt
WARNING Suspicious network activity.
ERROR Unauthorized access detected
INFO File accessed

Sample Output:

=== Log Summary ===
INFO : 2
ERROR : 2
WARNING : 1

8. Network Segmentation and Lateral Movement Prevention.

Pseudocode:

1. Start
2. Define network segments: Low-Trust, Employee, Application, Database and Internet.
3. Define the allowed connections between segments.
4. Read the source segment
5. Read the destination segment.
6. Check whether the connection is allowed by the access policy.
7. If allowed, display ACCESS ALLOWED.
8. Otherwise, display ACCESS DENIED.
9. Stop

Sample Input:

Enter source segment: Low-Trust
Enter destination segment: Database

Sample Output:

Source: Low-Trust
Destination: Database
Result: Access Denied